

AWS CloudWatch Lab: Collect EC2 VPC Flow Logs Using CloudWatch Logs

Summary:-

In this lab, **Amazon CloudWatch Logs** was configured to collect and monitor rejected network traffic for an **Amazon EC2** instance using **VPC Flow Logs**. A new EC2 instance was launched without allowing SSH access, and a VPC Flow Log was created for its network interface to capture only rejected traffic. A dedicated CloudWatch Log Group with a custom retention period was configured to store the logs. Finally, SSH connection attempts were made to the EC2 instance, and the rejected connection requests were successfully verified in CloudWatch Logs.

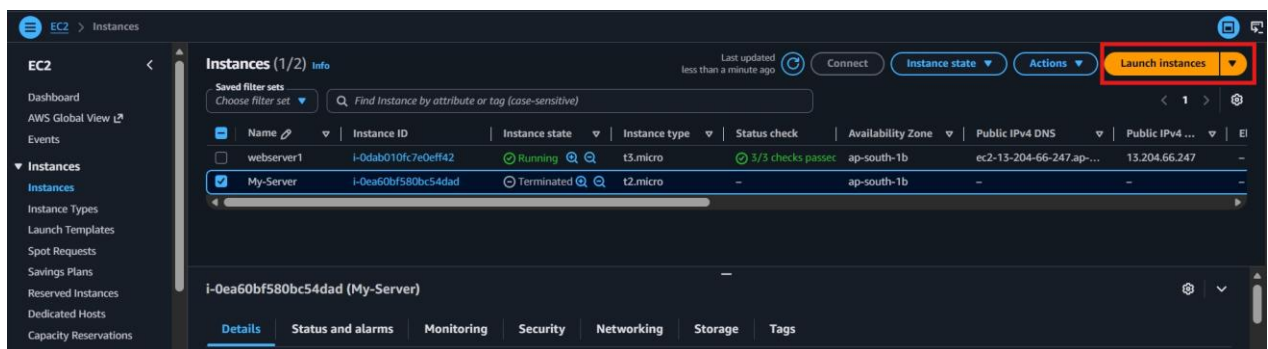
Let's Start:-

- **Amazon CloudWatch Logs** provides a centralized location to store and monitor logs generated by AWS resources.
- **VPC Flow Logs** capture information about the IP traffic flowing to and from network interfaces, subnets, or VPCs.
- In this lab, only **Rejected** network traffic is captured to monitor unauthorized SSH access attempts.

Lab Steps:-

Step 1 – Launch an EC2 Instance

- Sign in to the AWS Management Console.
- Navigate to Amazon EC2.
- Click Launch Instance.



- Enter an instance name (e.g., My Server).
- Select Amazon Linux as the AML.
- Choose a Free Tier eligible instance type (t2.micro or t3.micro).
- Select an existing Key Pair.
- Use the Default VPC.

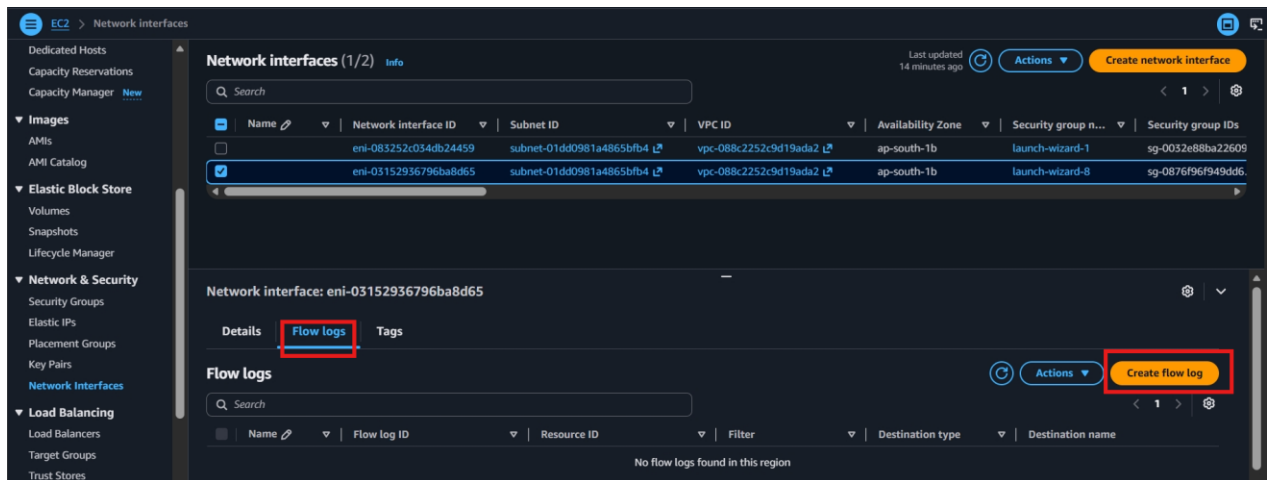
- Do not enable SSH (Port 22) in the security group.
- Click Launch Instance.

- Wait until the instance reaches the Running state.

Step 2 – Open the Network Interface

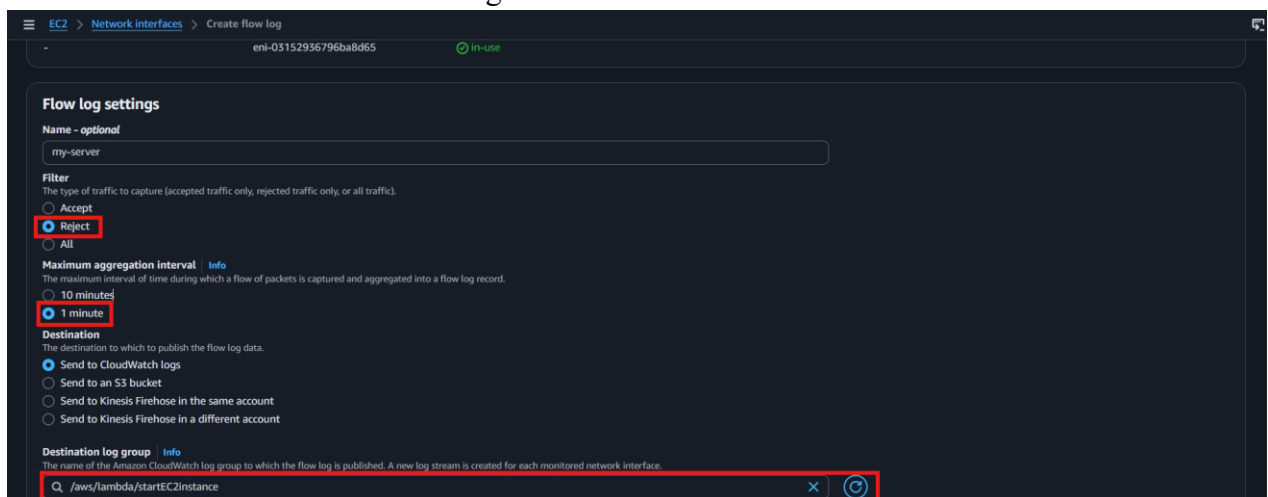
- Navigate to Amazon EC2.
- From the left navigation pane, select Network Interfaces.
- Locate the network interface attached to the EC2 instance.

- Select the network interface.
- Open the Flow Logs tab.
- Click Create Flow Log.



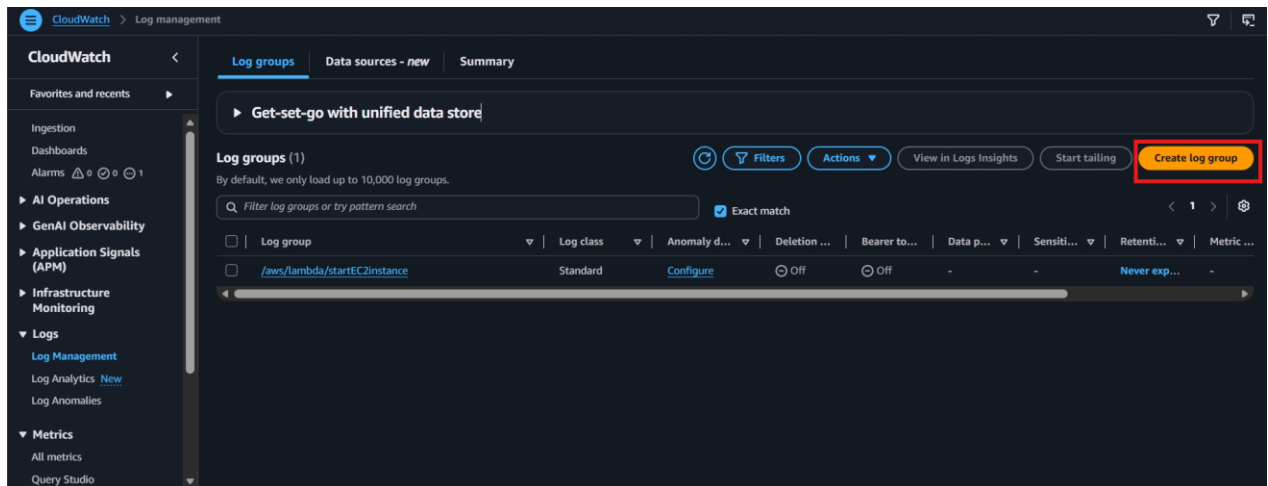
Step 3 – Configure the Flow Log

- Enter a Flow Log name.
- Set Filter to Reject.
- Set the Aggregation Interval to 1 Minute.
- Choose Send to CloudWatch Logs as the destination.

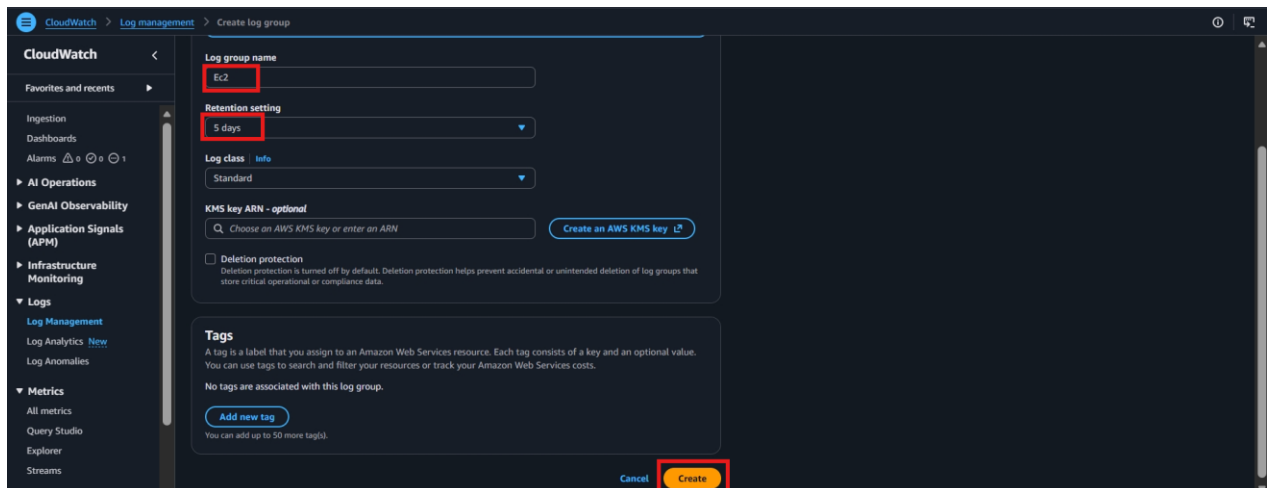


Step 4 – Create a CloudWatch Log Group

- Click Create New Log Group.

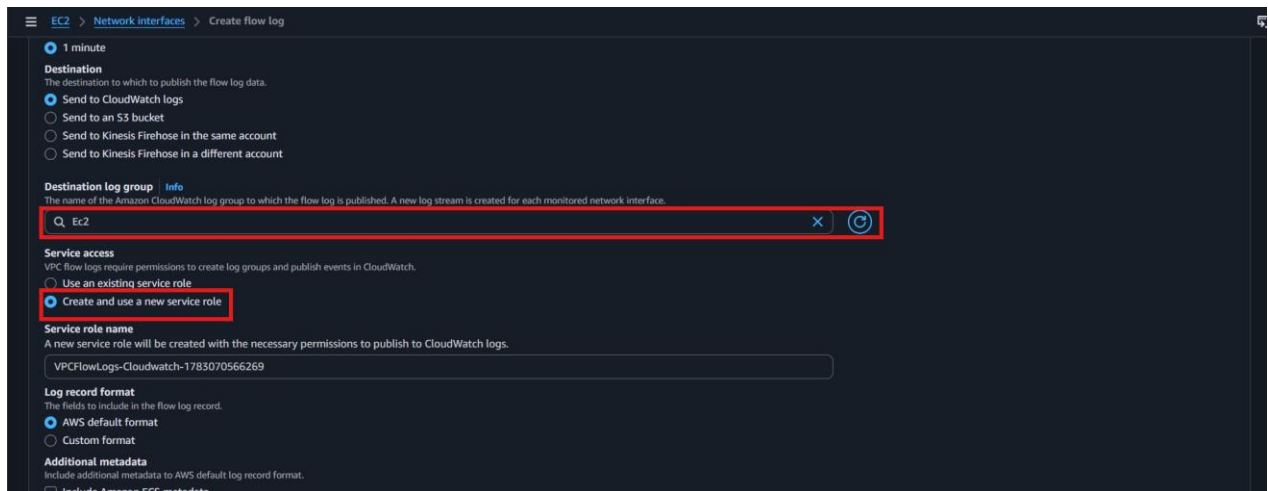


- Enter a log group name (e.g., EC2).
- Set the Retention Period to 5 Days.
- Keep the Log Class as Standard.
- Leave AWS KMS Encryption disabled.
- Click Create.



Step 5 – Complete the Flow Log Configuration

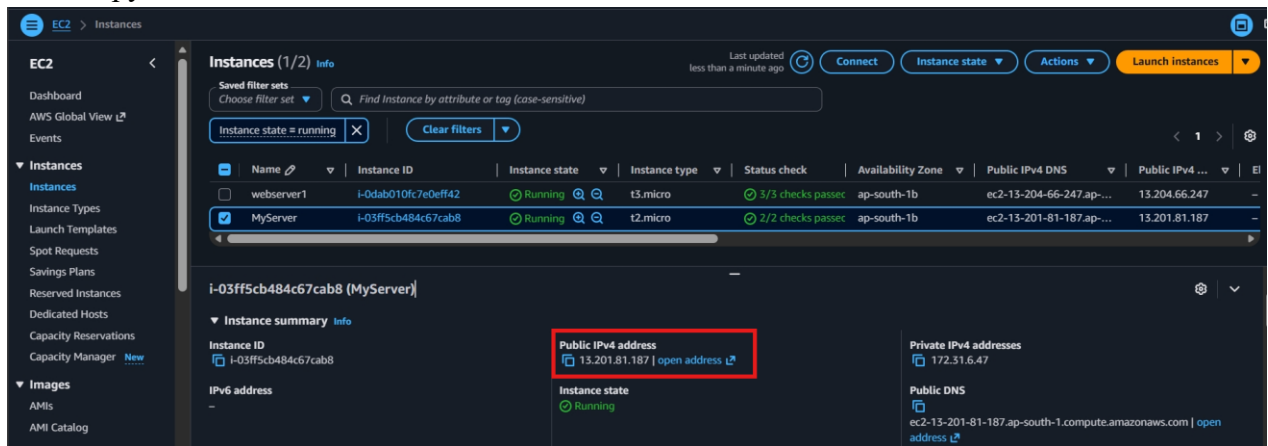
- Refresh the list of Log Groups.
- Select the newly created EC2 log group.
- Choose Create IAM Role Automatically.
- Keep the default log format.



- Click Create Flow Log.

Step 6 – Generate Rejected SSH Traffic

- Copy the Public IPv4 Address of the EC2 instance.



- Open Command Prompt or Terminal.
Enter: `ssh -i Testkey.pem ec2-user@< Public IPv4>`
- Attempt to connect to the instance using SSH.
- Verify that the SSH connection is rejected, since Port 22 is not allowed.

Step 7 – View the Flow Logs

- Navigate to Amazon CloudWatch.
- Open Log Groups.
- Select the EC2 log group.

Network interfaces (1/2) Info

Search

Network interface ID	Subnet ID	VPC ID	Availability Zone	Security group n...	Security group IDs	Inter
eni-03152936796ba8d65	subnet-01dd0981a4865bfb4	vpc-088c2252c9d19ada2	ap-south-1b	launch-wizard-8	sg-0876f96f949dd6...	Elast

Network interface: eni-03152936796ba8d65

Details Flow logs Tags

Flow logs (1/1)

Search

Name	Flow log ID	Resource ID	Filter	Destination type	Destination name
my-server	fl-0924ca38b6472e749	eni-03152936796ba8d65	REJECT	cloud-watch-logs	Ec2

- Open the available Log Stream.

CloudWatch > Log management > Ec2

ARN: arn:aws:logs:ap-south-1:386381158340:log-group:Ec2*

Creation time: 18 minutes ago

Retention: 5 days

Stored bytes: -

Metric filters: 0

Contributor Insights rules: -

KMS key ID: -

Deletion protection: Off

Data protection: -

Sensitive data count: -

Transformer: Configure

Anomaly detection: Configure

Bearer token authentication: Off

Syslog ingestion: Info

Log streams (1)

By default, we only load the most recent log streams.

Filter log streams or try prefix search

Exact match Show expired Info

Log stream Last event time

[eni-03152936796ba8d65-reject](#) 2026-07-03 09:25:23 (UTC)

- Wait a few moments if no logs appear immediately.

CloudWatch > Log management > Ec2 > eni-03152936796ba8d65-reject

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear 1m 30m 1h 12h Custom UTC timezone Display

Timestamp	Message
2026-07-03T09:38:56.000Z	2 386381158340 eni-03152936796ba8d65 64.62.156.76 172.31.6.47 46267 80 6 1 40 1783071536 1783071571 REJECT OK
2026-07-03T09:38:56.000Z	2 386381158340 eni-03152936796ba8d65 92.63.197.5 172.31.6.47 41261 64087 6 1 40 1783071536 1783071571 REJECT OK
2026-07-03T09:38:56.000Z	2 386381158340 eni-03152936796ba8d65 94.154.43.56 172.31.6.47 43317 8022 6 1 52 1783071536 1783071571 REJECT OK
2026-07-03T09:38:56.000Z	2 386381158340 eni-03152936796ba8d65 143.42.0.97 172.31.6.47 38760 2339 6 1 44 1783071536 1783071571 REJECT OK
2026-07-03T09:38:56.000Z	2 386381158340 eni-03152936796ba8d65 147.185.133.224 172.31.6.47 51433 45150 6 1 44 1783071536 1783071571 REJECT OK
2026-07-03T09:38:56.000Z	2 386381158340 eni-03152936796ba8d65 212.73.148.7 172.31.6.47 54484 4851 6 1 52 1783071536 1783071571 REJECT OK
2026-07-03T09:38:56.000Z	2 386381158340 eni-03152936796ba8d65 212.73.148.31 172.31.6.47 36130 2962 6 1 52 1783071536 1783071571 REJECT OK
2026-07-03T09:39:27.000Z	2 386381158340 eni-03152936796ba8d65 138.226.239.10 172.31.6.47 40733 29048 6 1 40 1783071567 1783071589 REJECT OK
2026-07-03T09:39:27.000Z	2 386381158340 eni-03152936796ba8d65 147.185.133.127 172.31.6.47 56118 45790 6 1 44 1783071567 1783071589 REJECT OK
2026-07-03T09:39:27.000Z	2 386381158340 eni-03152936796ba8d65 162.216.150.24 172.31.6.47 53047 32398 6 1 44 1783071567 1783071589 REJECT OK
2026-07-03T09:39:27.000Z	2 386381158340 eni-03152936796ba8d65 167.94.146.41 172.31.6.47 51795 32485 6 1 60 1783071567 1783071589 REJECT OK
2026-07-03T09:39:27.000Z	2 386381158340 eni-03152936796ba8d65 207.90.244.19 172.31.6.47 26280 880 6 1 44 1783071567 1783071589 REJECT OK
2026-07-03T09:39:27.000Z	2 386381158340 eni-03152936796ba8d65 212.73.148.25 172.31.6.47 35541 7492 6 1 52 1783071567 1783071589 REJECT OK

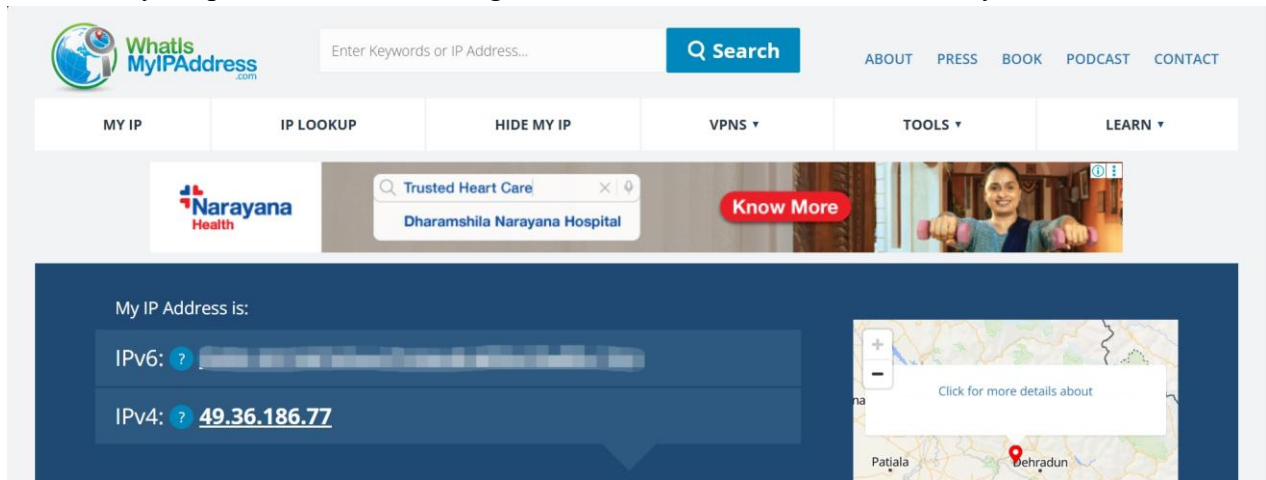
No newer events at this moment. Auto retry paused. [Resume](#)

[Back to top](#)

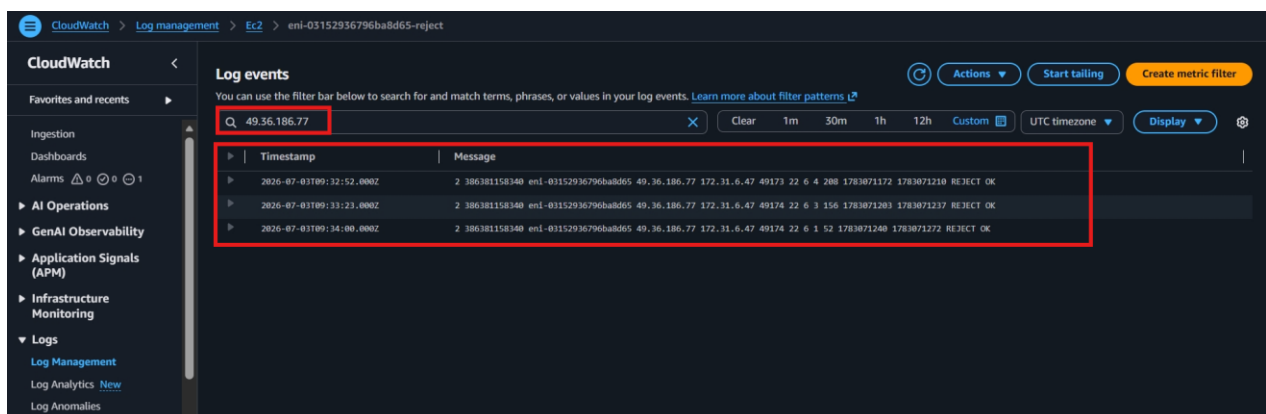
- Verify that rejected network traffic entries are displayed.

Step 8 – Search for Your Connection Attempt

- Find your public IP address using a browser search such as "What is my IP".

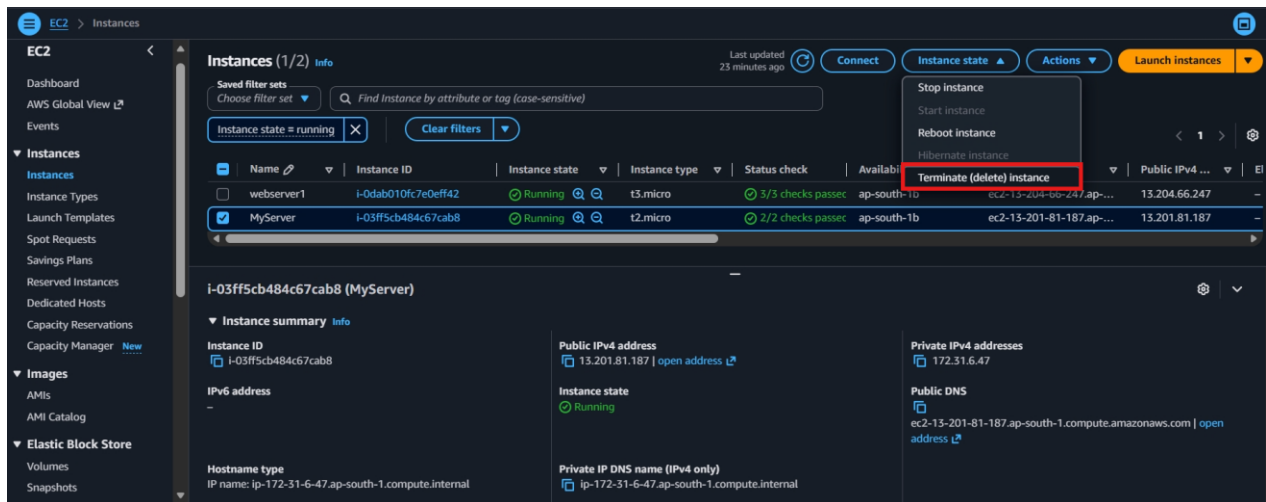


- Return to the CloudWatch Log Stream.
- Search for your public IP address.
- Verify that the connection attempt appears with Port 22 and the action marked as REJECT.



Step 9 – Clean Up Resources

- Delete the created Flow Log.
- Delete the CloudWatch Log Group if no longer required.
- Terminate the EC2 instance.



Verification

- The EC2 instance was launched successfully.
- SSH access was intentionally blocked by the security group.
- A VPC Flow Log was created for the EC2 network interface.
- A dedicated CloudWatch Log Group was configured successfully.
- Rejected SSH connection attempts were captured in CloudWatch Logs.
- The rejected entries displayed the source IP address, destination port (22), and REJECT action.
- All resources created during the lab were cleaned up successfully.